Game Theory (MGTHN0182)

Session 1: Strategic Games

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This lecture is based on the textbook ***A course in game theory*** by MJ Obsborn and A Rubinstein, MIT Press 1994.

“If people do not believe that mathematics is simple, it is only because they do not realize how complicated life is.”

— John von Neumann, 1947

Objectives

What this course **does** offer:

1. An introduction to Game Theory, a branch of economics.
2. A solid connection between Game Theory and its mathematical foundations.
3. A source of inspiration for modeling real-world strategic interactions.
4. A quite demanding course that enables you to use Game Theory for your thesis and beyond.

What this course does **not** offer:

1. A deep dive into the latest concepts.
2. It is not about playing games or designing games.
3. A 1-0-1 course on how to use game theory in a thesis.
4. A low-effort opportunity to get outstanding grades for attendance only.

Quantitative Skills

1. The course requires a certain aptitude in mathematics.
2. Session 2 might be perceived as difficult, and some parts of the proofs may be skipped at first.
3. Understand semantics, understand syntax.
4. I am here to support you.

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Battle of the Sexes Game

important

	soccer	ballet
soccer	2, 1	0, 0
ballet	0, 0	1, 2

Remark.

- *rows: player 1's* **actions**
- *columns: player 2's* **actions**
- *cells:* **payoff** *player 1, payoff player 2*

Battle of the Sexes Game

	soccer	ballet
soccer	2, 1	0, 0
ballet	0, 0	1, 2

Which activity does each player prefer?

Would they rather do their unpreferred event together or their preferred option alone?

What general context does this game capture?

Coordination Game

important

	Heilbronn	Munich
Heilbronn	2, 2	0, 0
Munich	0, 0	1, 1

Which city do the players prefer?

Which outcomes do you predict?

What general context does this game capture?

Prisoner's Dilemma Game

important

	don't confess	confess
don't confess	$-1, -1$	$-4, 0$
confess	$0, -4$	$-3, -3$

What is the best action for each player in this game?

What is the most desirable outcome if they could collaborate?

What general context does this game capture?

Prisoner's Dilemma Game: Different Positive Affine Payoff Representations

	don't confess	confess
don't confess	$-1, -1$	$-4, 0$
confess	$0, -4$	$-3, -3$

	don't confess	confess
don't confess	$3, 3$	$0, 4$
confess	$4, 0$	$1, 1$

Hawk-Dove Game

important

	dove	hawk
dove	3, 3	1, 4
hawk	4, 1	0, 0

What is the best action for both players in this game?

Will both behave dove-like?

What general context does this game capture?

Matching Pennies Game

important

	head	**tail**
head	1, 0	0, 1
tail	0, 1	1, 0

What is the best action for both players in this game?

What general context does this game capture?

Cournot Competition Game

important

There are two firms: $i \in \{1, 2\}$.

Each firm i decides on the quantity $q_i \geq 0$ it will offer to the market.

Firm i incurs costs of cq_i, where $0 < c < 1$.

The market-clearing price is $p(q_1 + q_2) = \max(0, 1 - q_1 - q_2)$.

The payoff functions are $u_i(q_1, q_2) := (p(q_1 + q_2) - c)\, q_i$.

What quantity will each firm supply?

Cournot Competition Game

Set of players	There are two firms: $i \in \{1, 2\}$.
Action sets	Each firm i decides on the quantity $q_i \geq 0$ it will offer to the market.
Payoff functions	Firm i incurs costs of cq_i, where $0 < c < 1$.
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Equilibrium	What quantity will each firm supply?

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Equilibrium	What quantity will each firm supply?

What general context does this game capture?

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important

Definition (Strategic Game)

A ***strategic game*** is a triple $\langle N, (A_i)_{i \in N}, (u_i)_{i \in N} \rangle$, where

- $N \subset \mathbb{N}$, N finite, called the ***set of players***,
- for each player $i \in N$, a nonempty set A_i, called the ***set of actions*** available to player i,
- for each player $i \in N$, $u_i : A \to \mathbb{R}$, where $A := \prod_{i \in N} A_i$, called the ***payoff functions***.

A strategic game is ***finite*** if A_i is finite for all $i \in N$.

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Remark.

1. *We can also write* $N = \{1, \ldots, n\}$.
2. *An element* $a = (a_i)_{i \in N} \in A$ *is called an* **action profile**.
3. *We write* (x_i) *instead of* $(x_i)_{i \in N}$, *when the qualifier* $i \in N$ *is obvious.*
4. *We define* $a_{-i} := (a_j)_{j \in N \setminus \{i\}}$ *for any action profiles* $a = (a_i)_{i \in N}$ *and* $i \in N$.
5. *We write* $(x_{-i}, x_i) := (x_1, x_2, \ldots, x_{i-1}, x_i, x_{i+1}, \ldots, x_n)$.
6. *von Neumann & Morgenstern (1944) say* **game in normal form** *to refer to a strategic game.*

Assumptions

We assume rationality

- Preferences are complete and transitive, and we assume a utility representation. Ordinal utility is invariant under strictly increasing transformations; expected-utility representations are unique up to positive affine transformations.
- Maximizing expected utility
- Perfect use of available knowledge
- Perfect backward induction (more about it later)
- Related concepts: hyper-rationality, bounded rationality, irrationality

Remark. *There are two common interpretations of a strategic form.*

1. *A one-shot game with rational players.*
2. *An instance of a sequence of infinitely often repeated independent games.*

Solution Concepts

Solution concepts

- Nash equilibrium
- subgame perfect equilibrium
- Bayesian Nash equilibrium
- ...

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Nash Equilibrium

important

Definition

A ***pure Nash equilibrium of a strategic game*** $\langle N, A, (u_i) \rangle$ is a profile $a^* \in A$ of actions with the property that for every player $i \in N$ we have

$$u_i \left(a^*_{-i}, a^*_i \right) \geq u_i \left(a^*_{-i}, a_i \right), \quad \forall a_i \in A_i.$$

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Remark.

- *If the action profile a is a pure Nash equilibrium of a strategic game, then no player can gain by unilaterally deviating from a_i.*
- *Canonical questions for each strategic game:*
  1. *Does the game have one Nash equilibrium?*
  2. *Is a Nash equilibrium unique?*
  3. *Is there an infinite number of Nash equilibria?*
- *Two common tasks:*
  1. *Identify all Nash equilibria*
  2. *Verify whether one candidate, $a \in A$, is a Nash equilibrium*

important

Definition

An action profile $a^P \in A$ is called ***Pareto efficient*** (or ***Pareto optimal***) if, whenever another profile $a \in A$ yields

$$u_j(a) > u_j(a^P) \quad \text{for some } j \in N,$$

then there exists at least one $i \in N$ with

$$u_i(a) < u_i(a^P).$$

Pareto Optimum

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Remark.

- *Pareto efficiency is not a solution concept in Game Theory.*
- *Pareto efficiency is an efficiency criterion in welfare economics.*
- *Pareto efficiency can be used in Game Theory to assess the welfare efficiency of an equilibrium.*

Battle of the Sexes Game Equilibria

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Prisoner's Dilemma Game Equilibrium

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Matching Pennies Game Equilibrium

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Cournot Competition Game Equilibrium

Set of players	There are two firms: $i \in \{1, 2\}$.
Action sets	Each firm i decides on the quantity $q_i \geq 0$ it will offer to the market.
Payoff functions	Firm i incurs costs of cq_i, where $0 < c < 1$. The market-clearing price is $p(q_1 + q_2) = \max(0, 1 - q_1 - q_2)$. The payoff functions are $u_i(q_1, q_2) := (p(q_1 + q_2) - c)\, q_i$.
Equilibrium	$q_1^* = q_2^* = \frac{1-c}{3}$ (Why?)
Symmetric Pareto optimum	$q_1^P = q_2^P = \frac{1-c}{4}$ (Why?)

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Definition (Power set, correspondence)

Let X, Y be nonempty sets.

1. By $\mathcal{P}(Y)$ we denote the ***power set*** of Y, which is the set that contains all subsets of Y.
2. A ***correspondence*** is a set-valued function, $f : X \to \mathcal{P}(Y)$, equivalently written as $f : X \rightrightarrows Y$.

important

Definition

Let $\langle N, A, (u_i) \rangle$ be a strategic game. The ***best-response correspondence*** of player i is the set-valued function

$$B_i : A_{-i} \to \mathcal{P}(A_i)$$

defined by

$$B_i(a_{-i}) = \left\{ a_i \in A_i \ : \ u_i(a_{-i}, a_i) \geq u_i(a_{-i}, a_i') \text{ for all } a_i' \in A_i \right\}.$$

Best responses

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Proposition

A strategy profile $a^* \in A$ is a **Nash equilibrium in pure strategies** *if and only if*

$$a_i^* \in B_i(a_{-i}^*) \quad \text{for all } i \in N.$$

Best responses

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A strategy profile $a^* \in A$ is a **Nash equilibrium in pure strategies** *if and only if*

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Proposition

Let $B : A \to \mathcal{P}(A)$ with $B(a) := \prod_{i \in N} B_i(a_{-i})$. A strategy profile $a^* \in A$ is a **Nash equilibrium** *in pure strategies if and only if*

$$a^* \in B(a^*).$$

Best Responses in the Cournot Game

Set of players	There are two firms: $i \in \{1, 2\}$.
Action sets	Each firm i decides on the quantity $q_i \geq 0$ it will offer to the market.
Payoff functions	The payoff functions are $u_i(q_1, q_2) := (p(q_1 + q_2) - c)\, q_i$.

Define,

$$b_i\left(q_{-i}\right) := \max\left\{0, \frac{1 - q_{-i} - c}{2}\right\}.$$

The **best response correspondences** are $B_i\left(q_{-i}\right) = \{b_i\left(q_{-i}\right)\}$. Alternatively, $B\left(q\right) = \{(b_1\left(q_2\right), b_2\left(q_1\right))\}$.

Best Response Functions

important

Proposition

Let $B_i(a_{-i}) = \{ a_i \in A_i : u_i(a_{-i}, a_i) \geq u_i(a_{-i}, a_i') \; \forall a_i' \in A_i \}$. If $|B_i(a_{-i})| = 1$ for all $i \in N$ and all $a_{-i} \in A_{-i}$, then there exists a function $b_i : A_{-i} \to A_i$ such that $B_i(a_{-i}) = \{b_i(a_{-i})\}$ for all $a_{-i} \in A_{-i}$. We call b_i the **best-response function**.

Remark.

1. *Strategic games do not have to have non-empty best-response correspondences (why?)*
2. *Strategic games with best-response correspondences do not have to have best-response functions (why?)*
3. *Strategic games with best-response functions have best-response correspondences (why?)*

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Topics in this course

0. Preliminaries
1. Games
2. Equilibria and Fixed-Point Theorems
3. Mixed Equilibrium
4. Rationalizability
5. Knowledge
6. Bayesian Games
7. Extensive Form Games
8. Repeated Games
9. Core
10. Stable Sets
11. Nash Bargaining

References

von Neumann, J. & Morgenstern, O. (1944), *Theory of Games and Economic Behavior*, Princeton University Press, Princeton, NJ.